**Code No: 121AE** **R15**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech I Year Examinations, March/April - 2023**
**ENGINEERING CHEMISTRY**
**(Common to CE, EEE, ME, ECE, CSE, IT, MIE, PTM)**

**Time: 3 hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART - A

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define molar conductance and its units. | [2] |
| b) | Why moderate temperature is used during electroplating. | [3] |
| c) | Define tactility and functionality of a monomer. | [2] |
| d) | Mention the characteristics of a good lubricant. | [3] |
| e) | Write the structure for EDTA. | [2] |
| f) | Distinguish between temporary and permanent hardness. | [3] |
| g) | What is meant by ignition temperature? | [2] |
| h) | Why should fuel for cars not evaporate too easily? | [3] |
| i) | Name two set systems which form eutectic mixtures. | [2] |
| j) | Name the type of adsorption with examples. | [3] |

## PART - B

**(50 Marks)**

2. a) Describe the construction of methanol-oxygen fuel cell with reactions.
b) Explain the mechanism of electro chemical corrosion. [5+5]

**OR**

3.a) Derive Nernst equation for single electrode potential. Mention its applications.
b) Write a brief note on sacrificial anodic impression. [5+5]

4.a) Explain in detail about injection molding method with a neat diagram.
b) Explain the preparation and applications of Butyl rubber and Thiokol rubber. [5+5]

**OR**

5.a) With the help of neat diagram write about thick film lubrication method.
b) Mention the applications of nanomaterials. [5+5]

6.a) Explain zeolite process for softening of water.
b) Write about reverse osmosis and its significance. [5+5]

**OR**

7.a) Discuss in detail about scale and sludge formation in water boilers. How is it controlled?
b) Write a brief notes on internal treatment of water. [5+5]

8.a) Describe the analysis of flue gas by orsat's apparatus.

b) Define calorific value. Distinguish between the gross and net calorific value. [5+5]

**OR**

9.a) Define knocking of a fuel. Discuss in detail how to improve the knocking characteristics of a fuel.

b) Explain the fixed –bed catalytic method to obtain gasoline. Give its mechanism. [5+5]

10.a) Draw a labelled diagram of water system and discuss the meta stable curve and principle of freeze drying.

b) Mention the classification of colloids with examples. [5+5]

**OR**

11.a) State Phase rule and discuss its applications to water system.

b) Distinguish between absorption and adsorption. [5+5]

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